PHYSICAL ACCESS AND CORRELATION GEOMETRY: THE HOLOGRAPHIC ALAYA–FIELD FRAMEWORK

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ABSTRACT

Physical access has a resolution that an observable algebra does not specify. We describe a reduction by its channel and distinguish its implementable effects from their linear span, their generated algebra and the observables stable under the dynamics. For qubit reductions we characterize equality of output entropy on every pure input. The nonzero branch is separated by every mixed input; the zero branch includes operationally incomparable channels with equal entropy on every input. A nine-probe finite-copy test bounds violations. We then connect physical implementation to geometric certification in one protocol: local unitary couplings reduce a fixed three-qubit GHZ state, further local collisions produce environmental records, and measurements on those records estimate the pair correlations. Calibrated readout errors and sampling errors propagate to confidence intervals for a fixed shortest-path metric. Two noise allocations with the same expected flip count produce non-isometric metrics, and a finite experiment separates their diameters without direct access to the source qubits. The reconstruction rule, source preparation and coupling architecture remain explicit inputs. The results show how finite measurement permissions, rather than an algebra label or exact entropy access, determine which correlation geometry can be established.

1 Introduction

Decoherence explains why interference between some degrees of freedom stops being visible [1]. Entanglement-based approaches to quantum gravity read geometric data from correlations between regions, most sharply through the Ryu–Takayanagi formula [2, 4, 3]. Both start after the system has been divided: into system, apparatus and environment in the first case, into boundary regions or tensor factors in the second. A generic Hilbert space carries no canonical division [5, 6, 7], and the observables that define a division can be chosen so that a given pure state has any Schmidt decomposition one likes [8]. Any geometry read from correlations therefore depends on how the state is reduced.

A reduction can preserve the algebra generated by accessible observables while changing the accuracy of every readout. This is the issue studied here. We first distinguish channel, effect set and stable structure (Section 2), then determine when qubit reductions are indistinguishable by entropy and how finite data detect a violation (Section 3). Section 4 connects a local interaction to an actual environmental record, a measured correlation and a certified finite geometry. Appendix A gives the factorization background and the precise typicality statement.

The geometry in the worked protocol is a metric on three sites under a declared rule. Its non-isometry is proved from the source dynamics and certified through the record dynamics. A Lorentzian metric requires further physical and reconstruction input. Keeping that distinction allows the finite experiment to answer a real part of the world-structure problem: which geometric differences survive the operations a reader can actually perform?

We call the source structure the Alaya-Field, a term taken from Yogācāra philosophy, where it names the storehouse consciousness. Here it is used only as a technical name for the triple $(\mathcal{H}_U, \mathcal{A}_U, |\Omega\rangle)$, with $\mathcal{A}_U$ a von Neumann algebra on $\mathcal{H}_U$ and $|\Omega\rangle$ cyclic and separating, so that the pair is in standard form. The point of the name is the absence of a canonical factorization, not a particular choice of state. In an algebraic quantum field theory $\mathcal{A}_U$ may be a Type III_1

factor [9]. Such a factor has no normal trace, so its states have no intrinsic density operators and no von Neumann trace entropy. The factorization, Haar-measure, partial-trace and entropy results below therefore use their own finite-dimensional representations. There a pure state of the full matrix algebra is cyclic but not separating, so these finite models do not realize every assumption placed on the infinite source structure, and none of their entropy statements is asserted in the Type III setting. The passage from entropy or connectivity data to a spacetime metric is a separate, model-dependent input throughout.

2 Coarse-graining structures, algebras and channels

2.1 Definition

An observable algebra does not by itself reinstate a tensor factorization: operators can act irreducibly on $\mathcal{H}_{\mathrm{total}}$, and in algebraic quantum field theory local algebras are assigned to regions without any global tensor product [9]. A coarse-graining here is meant to be realized by a physical interaction, not chosen by an agent.

Definition 2.1 (Coarse-graining structure)**.** *A coarse-graining structure is a pair* $\mathbf{c} = (\mathcal{A}_{\mathrm{sel},\mathbf{c}}, \Phi_{\mathbf{c}})$*, where*

1. $\mathcal{A}_{\mathrm{sel},\mathbf{c}} \subset \mathcal{B}(\mathcal{H}_U)$ *is a selected unital* $*$*-subalgebra used to define the effective representation. It is not identified with Paper E's probe-reachable effect system, pointer or record algebra, or environmental fixed set [26]; a physical model must state any map between them.*
2. *In the finite-dimensional results,* $\Phi_{\mathbf{c}}$ *is a completely positive trace-preserving (CPTP) map on states that implements an operational reduction; its Heisenberg dual* $\Phi_{\mathbf{c}}^*$ *is unital and completely positive. Partial trace over a specified factor and channels induced by a conditional expectation are examples. In a general von Neumann setting the corresponding object is a normal channel with a normal unital completely positive dual, and no entropy result below is claimed for that extension.*

No theorem guarantees a unique maximal observable algebra for a given physical system without additional structure. Several admissible algebras can coexist, and for a fixed algebra several channels can implement different losses of information, such as tracing out inaccessible degrees of freedom or effective decoherence. The two entries of $\mathbf{c}$ must therefore be related explicitly.

2.2 The algebra generated by a channel

For a channel $\Phi : \mathcal{B}(\mathcal{H})_* \to \mathcal{B}(\mathcal{K})_*$ put

$$\mathcal{S}_\Phi = \operatorname{ran} \Phi^*, \qquad \mathcal{A}_\Phi = C^*(\mathcal{S}_\Phi).$$

$\mathcal{S}_\Phi$ is a unital self-adjoint subspace, not necessarily closed under products. The effects that the channel actually reads out are $\Phi^*([0, I])$, which is in general smaller than the set of positive contractions in $\mathcal{S}_\Phi$. We call $\mathbf{c}$ weakly compatible if $\mathcal{S}_{\Phi_{\mathbf{c}}} \subseteq \mathcal{A}_{\mathrm{sel},\mathbf{c}}$ and minimally compatible if $\mathcal{A}_{\mathrm{sel},\mathbf{c}} = \mathcal{A}_{\Phi_{\mathbf{c}}}$.

Proposition 2.2 (Algebras and post-processing)**.**

1. *If two states agree on* $\mathcal{S}_\Phi$*, in particular if they agree on* $\mathcal{A}_\Phi$*, then their outputs under* Φ *coincide.*
2. *If* $\Phi_2 = \Lambda \circ \Phi_1$ *for a channel* Λ*, then* $\mathcal{S}_{\Phi_2} \subseteq \mathcal{S}_{\Phi_1}$ *and* $\mathcal{A}_{\Phi_2} \subseteq \mathcal{A}_{\Phi_1}$*.*
3. *The converse of (2) fails. For the depolarizing channels* $D_p(\rho) = p\rho + (1-p)I/2$ *on a qubit with* $0 < p < q \leq 1$*, both* $\mathcal{A}_{D_p}$ *and* $\mathcal{A}_{D_q}$ *equal the full matrix algebra,* $D_p = D_{p/q} \circ D_q$*, and* D_q *is not a post-processing of* D_p*.*

Proof. (1) For any output operator Y, $\mathrm{Tr}[\Phi(\rho)Y] - \mathrm{Tr}[\Phi(\sigma)Y] = \mathrm{Tr}[(\rho - \sigma)\Phi^*(Y)]$, and $\Phi^*(Y) \in \mathcal{S}_\Phi$. (2) $\Phi_2^* = \Phi_1^*\Lambda^*$, so the range of Φ_2^* lies in the range of Φ_1^*, and generated algebras inherit the inclusion. (3) Both maps are invertible on the operator space, so their dual ranges are everything. The identity $D_{p/q}(D_q(\rho)) = p\rho + (1-p)I/2$ is direct. Two orthogonal pure inputs have outputs at trace distance p under D_p and q under D_q; a channel cannot increase trace distance, so no Λ satisfies $\Lambda \circ D_p = D_q$. □

Two reductions with the same algebra can thus have different operational resolution. An algebra label is not a substitute for the channel and its allowed post-processing. If a conditional expectation is chosen as the implementation, that is an additional structure, not a synonym for every CPTP reduction.

2.3 Matching the dynamical and measurement interfaces

Paper B's commuting-coupling dynamics gives $\Phi_t(\rho)_{xy} = e^{-tD(x,y)}\rho_{xy}$ in the joint eigenbasis [25]. At finite t all multipliers are nonzero: $\mathcal{A}_{\Phi_t} = \mathcal{B}(\mathcal{H})$. Its block algebra $\mathcal{B}_\pi$ selected by a stability window is therefore an auxiliary description of persistent capability. Unless it is the full algebra, $(\mathcal{B}_\pi, \Phi_t)$ is not even weakly compatible in Definition 2.1. The minimally compatible finite-time pair is $(\mathcal{B}(\mathcal{H}), \Phi_t)$, supplemented by the window, tolerance and $\mathcal{B}_\pi$. In the long-time limit the channel is instead the pinching $\mathcal{E}_{\pi_\infty}$ and $(\mathcal{B}_{\pi_\infty}, \mathcal{E}_{\pi_\infty})$ is minimally compatible. The finite description and its limiting algebra answer different questions.

A measurement channel has a different output. In Paper E's pointer-controlled collision, the source remains a system while an environmental fragment receives a conditional state $|e_z\rangle$, $z = 0, 1$ [26]. Its record channel is

$$\Gamma(\rho) = \sum_{z=0}^{1} \mathrm{Tr}(P_z\rho)|e_z\rangle\langle e_z|, \qquad \Gamma^*(F) = \sum_z \langle e_z|F|e_z\rangle P_z.$$

For overlap $0 < r = |\langle e_0|e_1\rangle| < 1$, the dual range spans the pointer algebra, but the implementable effects omit its sharp projections. The source dephasing channel and Γ arise from the same unitary, although their ranges, outputs and uses differ. The geometric protocol below uses the reduction to set the correlations and Γ to read them. It never replaces $\Gamma^*([0, I])$ by all positive contractions of its span. Persistence is relative to the specified dynamics: a record effect fixed by the recording interaction need not be fixed by the preceding noise interaction.

2.4 Refinement

Definition 2.3 (Refinement)**.** *For coarse-graining structures $\mathbf{c}_1, \mathbf{c}_2$ write $\mathbf{c}_1 \succeq \mathbf{c}_2$ if there is a CPTP map Λ from states on the output space of $\Phi_{\mathbf{c}_1}$ to states on the output space of $\Phi_{\mathbf{c}_2}$ with $\Phi_{\mathbf{c}_2} = \Lambda \circ \Phi_{\mathbf{c}_1}$. Then $\mathbf{c}_2$ is a further coarse-graining of $\mathbf{c}_1$.*

Proposition 2.4 (Order on classes)**.** *The relation $\succeq$ is reflexive and transitive but not antisymmetric. Writing $\Phi \sim \Psi$ when each is a post-processing of the other, $\succeq$ induces a partial order on equivalence classes.*

Proof. Reflexivity uses $\Lambda = \mathrm{id}$, transitivity uses composition of channels. Two different replacer channels, which output fixed states $\omega_1 \neq \omega_2$, simulate each other by a further replacement, so antisymmetry fails. On classes the relation is well defined because composing with a simulating channel does not change membership, and antisymmetry holds by construction. □

If the allowed post-processings are restricted, for example to local, energy-bounded or specified control operations, the restricted class must be closed under composition before the same quotient can be formed. The comparison of channels by approximate post-processing has an operational characterization through success probabilities of discrimination tasks [14]; we use only exact post-processing.

2.5 Mutual information under refinement

Proposition 2.5 (Local and global post-processing)**.** *For channels Λ_A, Λ_B acting on A and B separately,*

$$I(A' : B')_{(\Lambda_A \otimes \Lambda_B)(\rho)} \leq I(A : B)_\rho.$$

A global CPTP post-processing can increase mutual information: the channel that replaces every two-qubit input by a Bell state takes $|00\rangle$, with $I = 0$, to a state with $I = 2\ln 2$.

Proof. $I(A : B)_\rho = D(\rho_{AB}\|\rho_A \otimes \rho_B)$, and $\Lambda_A \otimes \Lambda_B$ maps $\rho_A \otimes \rho_B$ to the product of the new marginals, so data processing for relative entropy gives the inequality. The Bell example is a direct computation. □

A requirement that mutual information decrease under refinement is therefore a theorem for local refinements and false for global ones. A geometric reading of mutual information must say which refinements it allows.

3 When do two reductions give the same entropies?

3.1 Generic separation

Theorem 3.1 (Generic separation for a fixed pair of channels)**.** *Let $\mathcal{H}$ and $\mathcal{K} = \mathcal{K}_A \otimes \mathcal{K}_{A^c}$ be finite dimensional, and let $\Phi_1 \neq \Phi_2$ be CPTP maps from states on $\mathcal{H}$ to states on $\mathcal{K}$. The pure inputs on which Φ_1 and Φ_2 agree form a proper*

real-algebraic subset of Haar measure zero. If, in addition, the real-analytic function

$$f_A(\Psi) := S(\mathrm{Tr}_{A^c}\,\Phi_1(|\Psi\rangle\langle\Psi|)) - S(\mathrm{Tr}_{A^c}\,\Phi_2(|\Psi\rangle\langle\Psi|))$$

is not identically zero on each connected component under consideration of the open set where both reduced outputs have full rank, then $f_A(\Psi) \neq 0$ for generic inputs in those components.

Proof. Pure-state projectors span the Hermitian operators, so $\Phi_1 - \Phi_2$ cannot vanish on all of them. The equality condition is polynomial in the real and imaginary parts of $|\Psi\rangle$, so its solution set is a proper real-algebraic subset and has Haar measure zero. On the full-rank domain the von Neumann entropy, and hence f_A, is real analytic, and a nonzero real-analytic function has a zero set of measure zero. □

The nondegeneracy hypothesis cannot be dropped: channels that differ by a unitary acting on A alone give different outputs with identical entropies on every input. Channel inequivalence alone therefore does not yield an entropic or geometric difference. The rest of this section decides, for qubit reductions, exactly when the hypothesis fails.

3.2 Qubit reductions: the criterion

A qubit channel acts on Bloch vectors as an affine map

$$\Phi_i :\ r \mapsto M_i r + t_i, \qquad i = 1, 2,$$

with M_i a real 3×3 matrix and $t_i \in \mathbb{R}^3$, subject to the complete-positivity conditions [15]. Put

$$A = M_1^T M_1 - M_2^T M_2, \qquad b = M_1^T t_1 - M_2^T t_2, \qquad c = \|t_1\|^2 - \|t_2\|^2.$$

The difference of the squared output Bloch lengths is

$$q(r) = \|M_1 r + t_1\|^2 - \|M_2 r + t_2\|^2 = r^T A r + 2b^T r + c. \tag{1}$$

The entropy of a qubit state is a strictly decreasing function of its Bloch length: $S = h\big((1+\|s\|)/2\big)$ with h the binary entropy in nats.

Theorem 3.2 (Equal entropy on all pure inputs)**.** *Φ_1 and Φ_2 give equal output entropy on every pure input if and only if there is a real number α with*

$$A = \alpha I, \qquad b = 0, \qquad c = -\alpha. \tag{2}$$

Proof. Equal entropy on pure inputs means $q(r) = 0$ on the unit sphere. Comparing $q(r)$ and $q(-r)$ gives $b^T r = 0$ for all unit r, so $b = 0$. A real symmetric quadratic form that is constant on the unit sphere is a multiple of the identity, so $A = \alpha I$ and then $c = -\alpha$. Conversely, (2) gives $q(r) = \alpha(\|r\|^2 - 1)$, which vanishes on the sphere. No rank condition on the outputs is needed, and pure outputs on the boundary are included. □

Together with positivity of the Choi matrices, (2) decides membership for any given pair. It is a decision rule, not yet a classification of all solutions into physically distinct normal forms.

3.3 The two branches

Proposition 3.3 (Zero branch)**.** *Let $X_i = [M_i \mid t_i]$ be the 3×4 matrix with t_i appended. If $\alpha = 0$, then $X_2 = OX_1$ for an orthogonal $O \in O(3)$. If $\mathrm{rank}\, X_1 \leq 2$, O can be chosen with $\det O = +1$, so Φ_2 is Φ_1 followed by a unitary conjugation of the output. If $\mathrm{rank}\, X_1 = 3$, O is unique; $\det O = +1$ again gives a unitary conjugation, and $\det O = -1$ gives a unitary conjugation composed with transposition.*

Proof. For $\alpha = 0$, (2) says exactly $X_1^T X_1 = X_2^T X_2$. Equal Gram matrices define an isometry between the column spaces of X_1 and X_2, which extends to an orthogonal map of $\mathbb{R}^3$. If the column space has dimension at most two, the extension can be composed with a reflection in an unused direction to make the determinant positive. Rotations of the Bloch ball are unitary conjugations, and the reflection $\mathrm{diag}(1, -1, 1)$ is transposition in the standard basis. □

Transposition is not completely positive, but both endpoints of a transposed pair can still be channels.

Example 3.4 (Equal entropies everywhere, no simulation either way)**.** *Take $M_1 = pI$, $M_2 = p\,\mathrm{diag}(1,-1,1)$ and $t_1 = t_2 = 0$. The first map is a depolarizing channel for $0 \le p \le 1$; by the conditions for unital qubit channels [16], the second is completely positive exactly when $0 \le p \le 1/3$. They give equal output entropy on every input, pure or mixed. For $0 < p \le 1/3$ neither is a post-processing of the other. Indeed both are invertible on the Bloch ball, so a post-processing Λ with $\Lambda \circ \Phi_1 = \Phi_2$ would have to act as $r \mapsto \mathrm{diag}(1,-1,1)r$ on an open set and hence everywhere; that map is transposition, whose normalized Choi matrix has smallest eigenvalue $-1/2$. The same argument excludes the reverse direction.*

Identical entropy functions on the whole input space therefore do not imply operational equivalence, and the two channels are incomparable in the order of Proposition 2.4.

Solutions with $\alpha \neq 0$ are not rare. Every pair of complementary qubit channels satisfies (2): if $V : \mathcal{H} \to \mathcal{K} \otimes \mathcal{E}$ is an isometry with $\mathcal{K}$ and $\mathcal{E}$ two-dimensional, $\Phi(\rho) = \mathrm{Tr}_{\mathcal{E}}\, V\rho V^\dagger$ and $\Phi^c(\rho) = \mathrm{Tr}_{\mathcal{K}}\, V\rho V^\dagger$, then for a pure input the joint output is pure and its two marginals have the same entropy. For amplitude damping with decay probability γ and its complementary channel, $\alpha = 1 - 2\gamma$, which is nonzero unless $\gamma = 1/2$. A family with full Choi rank is

$$\Phi_{a,b} : r \mapsto ar + b\hat{z}, \qquad \Phi_{b,a} : r \mapsto br + a\hat{z}, \qquad a, b > 0,\ a + b < 1.$$

$\Phi_{a,b}$ is a convex combination of the identity and the replacer with Bloch vector $b\hat{z}/(1-a)$, so it is CPTP, and likewise $\Phi_{b,a}$. Both give squared pure-output length $a^2 + b^2 + 2abr_z$, and $\alpha = a^2 - b^2$. For $a = .3$, $b = .2$ both normalized Choi matrices have smallest eigenvalue .125.

3.4 Mixed inputs expose the nonzero branch

Proposition 3.5 (Entropy gap on mixed inputs)**.** *For any two qubit channels and any input,*

$$|S_1(r) - S_2(r)| \ge \tfrac{1}{2}|q(r)|. \tag{3}$$

If (2) *holds, then $q(r) = \alpha(\|r\|^2 - 1)$ on the whole Bloch ball, so*

$$|S_1(r) - S_2(r)| \ge \frac{|\alpha|}{2}\big(1 - \|r\|^2\big), \tag{4}$$

and $\mathrm{sgn}(S_1 - S_2) = \mathrm{sgn}\,\alpha$ for every strictly mixed input.

Proof. Write the entropy as a function of the squared Bloch length, $f(x) = h\big((1+\sqrt{x})/2\big)$. Then $-f'(x) = \mathrm{artanh}(\sqrt{x})/(2\sqrt{x}) \ge \frac{1}{2}$, with the continuous extension at $x = 0$ and the integral limit at $x = 1$. The mean value theorem gives (3), and substituting (2) into (1) gives (4). A larger Bloch length means smaller entropy, which fixes the sign. □

When all pure inputs agree, a single strictly mixed input with equal entropies therefore forces $\alpha = 0$ and the orthogonal relation of Proposition 3.3. The argument uses only the one-to-one relation between qubit entropy and Bloch length, not a polynomial form of the entropy.

3.5 Probes and random inputs

Corollary 3.6 (Nine pure probes and one mixed probe)**.** *Let the probe set consist of the nine Bloch vectors*

$$\pm e_1,\ \pm e_2,\ \pm e_3,\ (e_1+e_2)/\sqrt{2},\ (e_1+e_3)/\sqrt{2},\ (e_2+e_3)/\sqrt{2}.$$

If the two channels give equal output entropies on these nine inputs, they give equal output entropies on every pure input. If they also agree at $r = 0$, they agree on every input.

Proof. The six axial probes give $b_i = 0$ and $A_{ii} + c = 0$; the three diagonal probes then give $A_{ij} = 0$ for $i \neq j$. This is (2). At $r = 0$, $q = c = -\alpha$, so agreement there gives $\alpha = 0$. Then q vanishes on the entire Bloch ball, hence the output lengths and entropies agree everywhere. □

The corollary concerns exact entropy values on nine inputs, not nine measurements. On the unit sphere $q(r) = r^T A_0 r + 2b^T r + d$ with $\alpha = \frac{1}{3}\,\mathrm{tr}\,A$, $A_0 = A - \alpha I$ and $d = c + \alpha$, which is linear in nine real parameters.

Proposition 3.7 (Random pure inputs)**.** *For Haar-random pure inputs,*

$$Q := \mathbb{E}\, q(r)^2 = \frac{2}{15}\|A_0\|_F^2 + \frac{4}{3}\|b\|^2 + d^2. \tag{5}$$

$Q = 0$ *exactly when* (2) *holds. If* $Q > 0$*, then* $\mathbb{E}(S_1 - S_2)^2 \geq Q/4$*, and for* $0 < \tau < 1$

$$\Pr\Big\{|S_1 - S_2| > \tfrac{1}{2}\sqrt{\tau Q}\Big\} \geq \Pr\{q^2 > \tau Q\} \geq \frac{(1-\tau)Q}{1-\tau Q}. \tag{6}$$

Proof. Use $\mathbb{E} r_i r_j = \delta_{ij}/3$ and $\mathbb{E} r_i r_j r_k r_l = (\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})/15$; odd moments and the cross terms with the traceless A_0 vanish. The second claim follows from (3). Both outputs are states, so $0 \leq q^2 \leq 1$; splitting $\mathbb{E}q^2 = Q$ at $q^2 = \tau Q$ gives $Q \leq \tau Q(1-P) + P$ with $P = \Pr\{q^2 > \tau Q\}$, which is (6). □

With n independent Haar inputs, the probability of missing the difference at the level $\frac{1}{2}\sqrt{\tau Q}$ is at most $(1 - p_\tau)^n$, where p_τ is the right side of (6). This assumes exact entropies; the next proposition removes that assumption. The average change of purity over pure inputs is also the basis of the unitarity measure of noise [17]; Eq. (5) concerns the difference between two channels rather than the coherence of one.

3.6 Finitely many copies

For a qubit, purity and entropy are both monotone functions of the Bloch length, so equal entropy is equivalent to equal purity, and purity can be measured directly. A SWAP test on two copies of a state σ returns the outcome 0 with probability $\frac{1}{2}\big(1 + \operatorname{Tr}\sigma^2\big)$ [18]. For a qubit with Bloch vector s this is $\frac{3}{4} + \frac{1}{4}\|s\|^2$. We assume that every channel use acts on a freshly prepared, known probe state, that the channels do not drift between uses, and that two output copies can be measured jointly.

Finite data cannot establish that the two entropy functions are exactly equal. They can separate equality from a violation of prescribed size, and they can bound how large a violation can be. Put

$$\kappa_9 = \frac{75 - 3\sqrt{465}}{8} = 1.28855\ldots.$$

Proposition 3.8 (Finite-copy test and confidence bound)**.** *For each of the nine probes* r_k *of Corollary 3.6, send* $2m$ *copies of the input through* Φ_1*, apply a SWAP test to each of the* m *output pairs, and do the same for* Φ_2*. Let* D_k *be the difference of the two observed frequencies of the outcome* 0*.*

1. *Given* $\gamma > 0$*, declare the channels different on pure inputs if* $\max_k |D_k| > \gamma/8$*. If* (2) *holds, a difference is declared with probability at most* $18\, e^{-m\gamma^2/64}$*. If* $\max_k |q(r_k)| \geq \gamma$*, the difference is missed with probability at most* $2\, e^{-m\gamma^2/64}$*.*

2. $\sum_k q(r_k)^2 \geq \kappa_9\, Q$*, and* κ_9 *is the largest constant with this property. Hence* $Q \geq Q_0 > 0$ *forces* $\max_k |q(r_k)| \geq (\kappa_9 Q_0/9)^{1/2}$*, and*
$$m \geq \frac{576\, \ln(18/\delta)}{\kappa_9\, Q_0}$$
output pairs per probe and channel keep both error probabilities below δ *for the hypotheses* $Q = 0$ *and* $Q \geq Q_0$*.*

3. *For every* $t > 0$*, with probability at least* $1 - 18\, e^{-mt^2}$*,*
$$Q \leq \frac{16}{\kappa_9} \sum_{k=1}^{9} \big(|D_k| + t\big)^2.$$

Proof. The expected value of D_k is $q(r_k)/4$. D_k is a sum of $2m$ independent Bernoulli variables with coefficients $\pm 1/m$, so Hoeffding's inequality [20] gives $\Pr\{|D_k - \mathbb{E}D_k| \geq t\} \leq 2e^{-mt^2}$. Under (2) every $\mathbb{E}D_k$ vanishes, and $t = \gamma/8$ with a union bound over the nine probes gives the first error bound. If $|q(r_k)| \geq \gamma$ for some k, then $|\mathbb{E}D_k| \geq \gamma/4$, and missing it requires $|D_k - \mathbb{E}D_k| \geq \gamma/8$, which gives the second.

The smallest generalized eigenvalue of the two quadratic forms $\sum_k q(r_k)^2$ and Q is κ_9, as computed in Appendix C. The largest of nine terms is at least their mean; setting $\gamma^2 = \kappa_9 Q_0/9$ and requiring $18e^{-m\gamma^2/64} \leq \delta$ gives item 2. For item 3, on the event that all nine deviations are at most t, which fails with probability at most $18e^{-mt^2}$, we have $|q(r_k)| = 4|\mathbb{E}D_k| \leq 4(|D_k| + t)$, and $Q \leq \kappa_9^{-1} \sum_k q(r_k)^2$. □

If the pure-input criterion is known to hold, the same argument at the mixed probe $r = 0$ tests the size of α: the expected difference there is $-\alpha/4$, and $m \geq 64\ln(2/\delta)/\alpha_0^2$ pairs per channel separate $\alpha = 0$ from $|\alpha| \geq \alpha_0$ with error at most δ in each direction. The pure test uses $18m$ channel uses per channel, $36m$ in all. The constants are conservative. Beyond qubits, purity and entropy are different functionals and this reduction no longer applies.

4 Connectivity and geometry

4.1 Mutual information as connectivity

To compute correlations, declare a subsystem or region representation of the channel output and its relation to the selected source algebra. The channel and state alone do not determine that representation. For output factors $A, B, \ldots$ in this representation, set

$$I_{\mathbf{c}}(A : B) = S_{\mathbf{c}}(A) + S_{\mathbf{c}}(B) - S_{\mathbf{c}}(AB),$$

with entropies computed after coarse-graining.

Definition 4.1 (Connectivity)**.** *For disjoint output factors A, B in the declared representation, the connectivity is $\mu_{\mathbf{c}}(A, B) := I_{\mathbf{c}}(A : B)$. It defines a weighted graph but is not a distance: large mutual information denotes strong correlation rather than large separation, and the metric axioms generally fail.*

A state-parameter metric, such as the SLD Fisher metric in Appendix B, is a different construction. Neither it nor the graph weights alone supply a Lorentzian metric.

4.2 A finite reconstruction rule

For a declared family of finite-dimensional subsystems, a positive connected correlation graph supports a shortest-path metric once an edge-length rule has been chosen. We use $\ell_e = 1/I_e$ on edges with $I_e > 0$. An edge with $I_e = 0$ is absent; distinct connected components then have infinite separation under this rule. Mutual information itself is not a distance. The rule assigns longer distances to weaker correlations, but its physical interpretation depends on the model.

A positive lower bound on the used mutual informations makes this rule stable. No assumption of exponential decay is needed for this finite statement. Exponential decay relative to a pre-existing locality structure can help construct a spatial interpretation [24, 23]; it would supply an additional input rather than derive that locality from the graph.

Proposition 4.2 (Stability of a graph metric)**.** *On the complete graph with v vertices, give the edge e the length $1/I_e$ and use the shortest-path metric d. If every $I_e \geq m > 0$ and the perturbed weights satisfy $|I'_e - I_e| \leq \varepsilon < m$, then*

$$\max_{i,j} |d'_{ij} - d_{ij}| \leq (v-1)\frac{\varepsilon}{m(m-\varepsilon)}.$$

Proof. For each edge, $|1/I'_e - 1/I_e| = |I_e - I'_e|/(I_e I'_e) \leq \varepsilon/(m(m-\varepsilon))$. A shortest path with positive weights uses at most $v-1$ edges. Comparing the new shortest path with the old lengths, and the old shortest path with the new lengths, bounds the difference in both directions. □

4.3 From entropy data to a spacetime metric

In holography, the entropy of a boundary region equals the area of an extremal surface in Planck units [4], and the connectivity of spacetime has been argued to track the pattern of entanglement [3]. In tensor-network models, graph distance in an entanglement graph can approximate continuum distance [22]. These are reconstruction dictionaries for particular classes of states.

For a comparison of spacetime geometries, a dictionary $\mathfrak{G}$ must also state which entropy profiles it identifies. A sufficient condition is

$$\mathfrak{G}[S_1] \simeq_{\text{diff}} \mathfrak{G}[S_2] \implies S_1 \simeq_{\mathfrak{G}} S_2.$$

Its contrapositive says that separated profiles produce separated geometries. This is a requirement on the dictionary, not a derivation of geometry from a channel. The next result replaces that requirement by a fixed finite rule for which the distances and their measurement uncertainties can be computed.

4.4 One state, one operation family, one rule

Proposition 4.3 (Non-isometric finite geometries)**.** *Fix the three-qubit GHZ state. Allow only local bit-flip channels*

$$\Phi_{\mathbf{p}} = \bigotimes_{i=1}^{3} \big[(1-p_i)\,\mathrm{id} + p_i\,\mathrm{Ad}_{X_i}\big], \qquad 0 < p_i < \tfrac{1}{2},$$

which are generated by local bit-flip semigroups with $p_i = (1-e^{-2\gamma_i t})/2$. Fix the reconstruction rule: on the complete graph of the three qubits, the edge $\{i,j\}$ has length $1/I_{ij}$, and distances are shortest-path lengths. Then

$$I_{ij} = \ln 2 - h\Big(\frac{1+c_{ij}}{2}\Big) > 0, \qquad c_{ij} = (1-2p_i)(1-2p_j),$$

and different flip probabilities can give metric spaces that are not isometric under any relabelling of the vertices:

flip probabilities	*mutual informations of the three edges*	*diameter*
$(.1, .1, .1)$	$(.221753694,\ .221753694,\ .221753694)$	4.509507747
$(.05, .1, .2)$	$(.288183695,\ .153870839,\ .120090263)$	8.327069751
$(.05, .1, .15)$	$(.288183695,\ .214256042,\ .166239219)$	6.015427679

The first and third rows have the same expected number of applied flips, $\sum_i p_i = .3$, when each channel is implemented by applying X_i with probability p_i.

Proof. Every two-qubit marginal of the GHZ state is an equal mixture of $|00\rangle$ and $|11\rangle$. Local flips leave both bits uniform and make them agree with probability $(1+c_{ij})/2$, so $I_{ij} = 2\ln 2 - \big(\ln 2 + h((1+c_{ij})/2)\big)$. All edge lengths are finite and positive, so the rule defines a metric. The diameters differ, so no relabelling makes the spaces isometric. The numbers were also computed from the full eight-dimensional density matrix with the channels applied and partial traces taken; the analytic formula agrees to better than 3×10^{-16}. □

Every entry is fixed in advance: the state, the class of allowed operations, the interpretation of the vertices and the reconstruction rule. Nothing about the target geometry is written into the channels. The local flip channels are invertible on the operator space, so all of them generate the full matrix algebra; the algebra label is the same in every row while the geometry is not, which is Proposition 2.2(3) in geometric form. The equal flip budget in rows one and three shows that the difference comes from how a fixed resource is distributed among the qubits, not from its total. This is a one-step budget in a stated implementation; it does not claim equal cost on all hardware or under continuous-time noise. By Proposition 4.2, perturbing the three mutual informations by ε changes each distance by at most $2\varepsilon/(m(m-\varepsilon))$ with m the smallest mutual information. The model is a finite existence result for a fixed rule, not a derivation of the rule from mutual information and not a spacetime.

4.5 The reduction is a local interaction

For each source qubit prepare a noise ancilla N_i in $|0\rangle$ and apply

$$U_i(\theta_i) = e^{-i\theta_i X_i \otimes Y_{N_i}}, \qquad U_i(\theta_i)|\psi\rangle|0\rangle = \cos\theta_i|\psi\rangle|0\rangle + \sin\theta_i X_i|\psi\rangle|1\rangle.$$

Discarding N_i realizes exactly the local flip channel with $p_i = \sin^2\theta_i$. One possible interaction Hamiltonian is $\hbar g_i X_i \otimes Y_{N_i}$ applied for θ_i/g_i. The angles, tensor factors and timing are supplied by the apparatus. They do not encode a target metric.

Next attach, at each site, a fragment of k_i fresh memory qubits. Use Paper E's collisions $P_{0,i} \otimes I + P_{1,i} \otimes R_{i,a}$, with real conditional overlaps $c_{i,a}$, so the fragment overlap is $r_i = \prod_{a=1}^{k_i} c_{i,a}$. These collisions dephase the source in the Z basis. Every two-site marginal of the noisy GHZ state is already diagonal, so its probabilities and mutual information are unchanged by recording. The post-recording source therefore has the same graph metric as before, although its three-site coherence may have changed. This is preservation of the data used by this rule, not preservation of the entire source state.

The noise gate ends before recording begins. This sequencing and its switching resources matter: the flip couples through X, while the recorder uses Z. Their common fixed algebra is only the scalars. Indeed the composed single-site source channel sends a Bloch vector to $(r_i x, r_i(1-2p_i)y, (1-2p_i)z)$; for $0 < p_i < 1/2$ and $0 < r_i < 1$ it has no fixed nonidentity direction, although its linear range is still the full matrix algebra. The record channel nevertheless

reads the final pointer probabilities with nonzero resolution. This supplies an actual finite-access example beyond an exact stable-algebra label.

Tracing out the source after the collisions gives the exact joint fragment state

$$\rho_{\mathrm{rec}} = \sum_{\mathbf{z}} P_{\mathbf{p}}(\mathbf{z}) \bigotimes_{i=1}^{3} |e_{z_i}^{(i)}\rangle\langle e_{z_i}^{(i)}|, \tag{7}$$

where $z_i = \pm 1$ labels the local pointer and $P_{\mathbf{p}}$ is the actual diagonal distribution of the noisy GHZ state. Thus conditional product structure follows from the unitary couplings and fresh product memories; it is not an extra assumption about independent answers. Each site has equal pointer priors.

For $r_i < 1$, the equal-prior Helstrom measurement on fragment i has a symmetric binary error probability $\epsilon_i = (1 - \sqrt{1 - r_i^2})/2$ [19, 26]. To see the symmetry directly, choose real vectors $e_0 = (1, 0)$ and $e_1 = (r, \sqrt{1 - r^2})$. The sign of $|e_0\rangle\langle e_0| - |e_1\rangle\langle e_1|$ has expectations $+\sqrt{1 - r^2}$ and $-\sqrt{1 - r^2}$ in the two states. Writing $Y_i = \pm 1$ for the actual outcome and $v_i = 1 - 2\epsilon_i = \sqrt{1 - r_i^2}$ therefore gives

$$\mathbb{E}(Y_i \mid \mathbf{z}) = v_i z_i, \qquad \mathbb{E}(Y_i Y_j) = v_i v_j c_{ij}. \tag{8}$$

Local measurement on (7) gives these identities even though the records are correlated before conditioning.

4.6 Finite records certify the geometric difference

Let $f(c) = \ln 2 - h((1 + c)/2)$, $-1 \le c \le 1$. Consider K noise configurations, with independently prepared source states and fresh ancillas in each run. For every configuration collect n triples (Y_1, Y_2, Y_3) from the local measurements above. Calibration supplies estimates $\widehat{v}_i \in [0, 1]$ with $\widehat{v}_i \ge \nu > 0$ and $|\widehat{v}_i - v_i| \le \xi$. The estimate

$$\widehat{c}_{ij} = \mathrm{clip}_{[-1,1]} \frac{n^{-1} \sum_{s=1}^{n} Y_i^{(s)} Y_j^{(s)}}{\widehat{v}_i \widehat{v}_j}$$

uses the measurement outcomes, not hidden pointer values.

Theorem 4.4 (Record-to-metric confidence certificate)**.** *Suppose the calibration event has failure probability at most δ_{cal}, and the record law is* (7) *with the stipulated local measurements. For every $\tau > 0$, with probability at least $1 - \delta_{\mathrm{cal}} - 6Ke^{-n\tau^2/2}$, all $3K$ correlations satisfy*

$$|\widehat{c}_{ij} - c_{ij}| \le e_c := \frac{\tau + 2\xi}{\nu^2}. \tag{9}$$

Intersect each interval $[\widehat{c}_{ij} - e_c, \widehat{c}_{ij} + e_c]$ with $[-1, 1]$ and minimize and maximize f on it, obtaining I_{ij}^-, I_{ij}^+. If every $I_{ij}^- > 0$, form shortest-path metrics d^- and d^+ from edge lengths $1/I_{ij}^+$ and $1/I_{ij}^-$. The true metric obeys $d^- \le d \le d^+$ entrywise, and its diameter lies in $[\operatorname{diam} d^-, \operatorname{diam} d^+]$. Disjoint diameter intervals for two configurations certify non-isometry on the same confidence event.

For a sample-size guarantee, if true and estimated correlations obey $|c_{ij}|, |\widehat{c}_{ij}| \le b < 1$, set $e_I = \operatorname{artanh}(b) e_c$. When $I_{ij} \ge \mu > e_I$,

$$\|\widehat{d} - d\|_\infty \le R := \frac{2e_I}{\mu(\mu - e_I)}. \tag{10}$$

For two configurations with true diameter gap $\Delta > 4R$, the intervals $[\operatorname{diam} \widehat{d} - R, \operatorname{diam} \widehat{d} + R]$ are disjoint on that event.

Proof. Each $Y_i Y_j \in [-1, 1]$ has expectation $v_i v_j c_{ij}$. Hoeffding's inequality gives a two-sided tail $2e^{-n\tau^2/2}$; a union bound covers the $3K$ edge estimates. The three edges within a run need not be independent. On the calibration event, $|v_i v_j - \widehat{v}_i \widehat{v}_j| \le 2\xi$, by telescoping and the bounds $v_i, \widehat{v}_i \le 1$. Dividing by $\widehat{v}_i \widehat{v}_j \ge \nu^2$, using $|c_{ij}| \le 1$, and clipping cannot increase the error, proving (9). The interval construction contains the true mutual informations. Shortest-path lengths and their diameter are monotone in positive edge lengths, proving the metric enclosure. Finally $f'(c) = \operatorname{artanh} c$, so $|\widehat{I} - I| \le e_I$. Proposition 4.2, with three vertices, gives (10). Each estimated diameter differs from its true value by at most R, so the estimated gap is at least $\Delta - 2R$; two radius-R intervals are disjoint when $\Delta > 4R$. □

For the equal-flip-count configurations $(.1, .1, .1)$ and $(.05, .1, .15)$, take one recording qubit per site with overlap $r_i = .2$. Then $v_i = \sqrt{.96}$ and the correlations lie between .56 and .72. With exactly specified v_i, use $\nu = \sqrt{.96}$, $\xi = 0$, $\tau = .003$, $b = .8$, $\mu = .16$ and $\delta = .05$. The choice

$$n = \left\lceil \frac{2\ln(12/\delta)}{\tau^2} \right\rceil = 1\,217\,920$$

runs per configuration gives $e_c = .003125$, $e_I = .0034332$ and $R < .275$. The true diameter gap is $1.505919931 > 4R$. Thus a finite record-only experiment separates the two geometries with confidence at least .95, under the stated source and apparatus model. The sample bound is conservative and does not claim optimal tomography.

A known unitary specification is not free empirical calibration. If the v_i are unknown, prepare reference sources in $Z = +1$ and apply the same record couplings and readers. Each reference outcome has mean v_i. For N independent reference runs per site, clipping the empirical mean to $[0, 1]$ yields

$$\Pr\{\max_i |\widehat{v}_i - v_i| > \xi\} \le 6e^{-N\xi^2/2}$$

for three shared readers; separately calibrated readers for K configurations replace 6 by $6K$. Thus $N \ge \lceil 2\ln(6/\delta_{\mathrm{cal}})/\xi^2 \rceil$ per site suffices. Reject calibrations below ν. For the same two geometries, take $\xi = .001$, $\delta_{\mathrm{cal}} = .025$, $\nu = .978$ and $\tau = .002$. Then $N = 10\,961\,278$ reference runs per site and $n = 3\,086\,894$ main runs per configuration give total failure at most .05, $e_c < .004182$, $R < .36955$ and $4R < 1.4782 < \Delta$. On the calibration event the true $v_i = \sqrt{.96}$ ensures $\widehat{v}_i > \nu$; the correlation bound also stays inside $b = .8$. The reference budget is therefore $3N$ additional preparations and readouts, not part of the $2n$ main runs. Equation (9) includes their residual error. Prepared reference pointers, their basis labels and the stability of the reader between calibration and use are additional physical resources.

The comparison uses $2n$ preparations of the same GHZ source, $6n$ fresh noise ancillas, $6n$ fresh recording qubits, $6n$ local binary readouts and communication of the outcomes to a comparator, plus any calibration runs. The expected noise-flip count is .3 per source in both configurations; their recording and readout schedule is identical. Preparation, switching work, storage, communication and renewal do not vanish because the counts agree. No heat or total-energy equality follows from the flip count, and this finite protocol supplies no reservoir reset.

4.7 Limits of record inversion

At $r_i = 1$ the record states coincide and $v_i = 0$: algebraic inversion of their span supplies no source information. Even for $v_i > 0$ the error grows as the inverse product visibility. A narrow effect set can span the pointer algebra while requiring too many trials for the available budget.

Calibration of individual error rates is also insufficient when the readers share an unmodelled error. Let a common random sign B multiply all otherwise ideal outcomes, with $\mathbb{E}B = v$. Each local reference test then reports visibility v, while the pair products are unchanged, since $B^2 = 1$. Dividing by v^2 overestimates the true correlations. A common inversion has $v = -1$ and is invisible to agreement. Joint reference tests, conditional error modelling or a verified local apparatus are needed to exclude such contamination. The confidence certificate is conditional on the joint record law, rather than an inference of that law from single-site calibration.

A spacetime reconstruction must solve a further problem: finite measurements determine a profile within an error region, not an exact profile. The relevant dictionary must be stable on that region and must quotient its coordinate freedom. Entropy equality from Section 3 supplies a genuine obstruction, including operationally different channels with identical entropy profiles. More accurate readout cannot repair a dictionary's failure to distinguish such profiles.

Conjecture 4.5 (Finite permissions approaching geometric access)**.** *In a source family with an independently justified local reconstruction dictionary, histories that enlarge the available record fragments and calibrate their readout can make the inferred geometry converge while the generated observable algebra remains constant. The required growth of visibility, preparation count and dictionary stability bounds must jointly dominate sampling and model error.*

The finite certificate supplies the mechanism and the costs of this candidate. For E edges with visibility at least ν, error in the correlations scales as $\nu^{-2}\sqrt{\ln(2E/\delta)/n}$ before calibration error. If reconstructed edge mutual informations have lower bound μ, distance error carries another factor of order μ^{-2} times the number of edges in a shortest path. Consequently, a growing graph or an approaching zero-information edge can defeat convergence even while all fixed-algebra labels are unchanged. The conjecture would fail in a proposed model if its dictionary is unstable, if common error survives calibration, or if its available budget cannot overcome these factors. This is a route from finite permissions to geometric access, not an assumption that a continuum or its dynamics is already generated.

The spatial/eigenmode example in Appendix B illustrates dependence of connectivity on factorization. Unlike Proposition 4.3, its empty eigenmode graph has no finite reconstructed paths.

4.8 Not a change of coordinates

A diffeomorphism acts within a fixed algebra of observables and leaves unchanged what can be measured. A change of coarse-graining changes the accessible observables, which correlations are visible and how subsystems are defined. Algebraic difference alone does not prove that two reconstructed metrics are not diffeomorphic, because a dictionary may discard it; the conclusion needs a separating dictionary or an explicit computation such as Proposition 4.3.

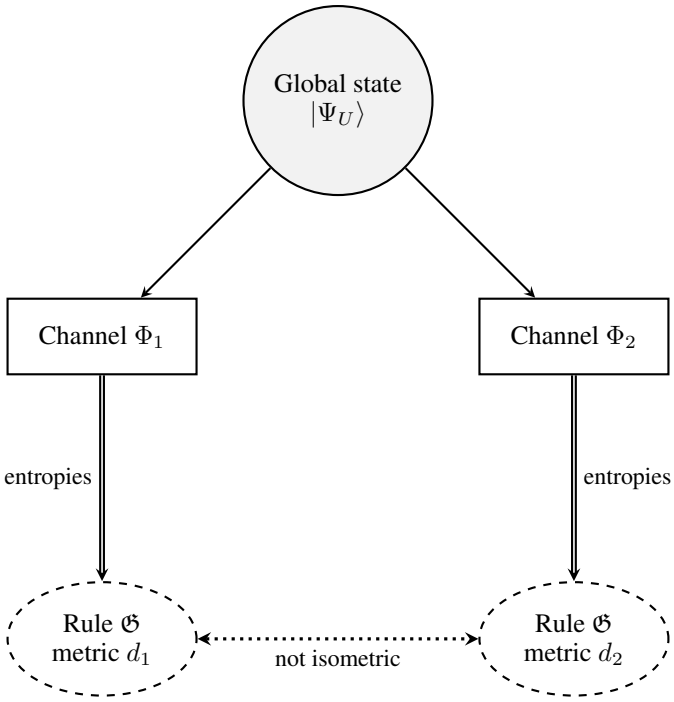


Figure 1: One global state is reduced by two channels from the same allowed family. The same reconstruction rule turns their entropy data into metrics. Section 3 decides when the entropy data differ; Proposition 4.3 gives a case where the metrics are not isometric.

5 Discussion

5.1 Relation to other frameworks

QBism treats the quantum state as an agent's degree of belief. Here a single objective global state is assumed throughout, and coarse-grainings are realized by physical interactions. The only point of contact is that no subsystem decomposition is privileged. Decoherence and many-worlds accounts explain classical behaviour and branching relative to a fixed split into system, apparatus and environment. The present framework leaves that analysis unchanged and asks where the split comes from. The closest relative is algebraic quantum field theory, where algebras are primary and states are functionals on them [9], and the closest technical precursor is the result that tensor-product structures are induced by the accessible observables [6]. What the present paper adds to that line is the channel level: a reduction has a resolution that its algebra does not record (Proposition 2.2), entropy-based readings have exact blind spots (Theorem 3.2, Example 3.4), those blind spots can be tested with finite resources (Proposition 3.8), and a fixed rule can turn a different distribution of the same noise budget into a different finite geometry (Proposition 4.3). The affine form of qubit channels, the complete-positivity conditions and the comparison of channels by post-processing are known tools [15, 16, 14]; the criterion, the branch analysis and the tests are derived here, and a full comparison with the literature on channels with equal output spectra is still to be done.

5.2 Series interfaces and the world-structure question

The protocol connects three distinct responsibilities. Paper B derives a reduction and its persistent structures from specified couplings. Its window theorem selects a block partition when within-block and between-block rates permit one; this does not make the finite channel's full range equal to that block algebra. Paper E derives the fragment's implementable effects and its public readout properties from a collision. Here their composition reaches a certified geometric difference. The source-channel algebra can stay fixed while recording permission and the attainable precision change. Imprint's cross-scale work develops the related question of which reading is worth its cost [27].

The coupling operators, factorization into sites, source preparation and reconstruction rule are still inputs. The new connection derives what happens between those inputs and a finite reader's geometric conclusion; it does not derive the architecture from an unstructured Hilbert space. Histories that alter actual coupling or record access, as studied in the subject and maintenance lines of the trilogy, can change this capability. A claim that they also generate locality or a spacetime dictionary requires its own dynamics and bridge, including the error conditions of Conjecture 4.5.

The finite models use trace entropy, whereas the infinite source description introduced above does not. Neither the SLD Fisher metric nor a shortest-path metric is identified with Lorentzian spacetime. The task is to determine which operational structures can constrain a physical geometric reading, and which geometric claims fail because a finite context cannot distinguish the data.

6 Conclusion

A physical reduction carries more information than its generated algebra. Two reductions may share the algebra and differ in resolution; they may agree on every entropy and remain operationally incomparable. For qubits these possibilities are decided by a quadratic criterion, and finite-copy tests bound its violation.

The worked protocol carries the argument through actual operations. A local unitary produces the source noise, local collisions create environmental records without changing the pair data used by the rule, and the measured records certify non-isometric shortest-path metrics. Noise allocation changes geometry at equal expected flip count. The finite sample guarantee includes imperfect readout, calibration and a common-error failure mechanism, so the conclusion is attainable by the declared readers.

The resulting dependence is physical and quantitative within the model: changing an implemented coupling changes a state, its recorded correlations and the reconstructed metric. Explaining the coupling architecture, deriving a spacetime dictionary and its continuum dynamics remain the larger world-structure problem. Conjecture 4.5 gives a mechanism and failure conditions for one part of that extension. No empirical departure from quantum theory is claimed; the protocol itself gives testable statements.

A No canonical tensor factorization

For a Hilbert space $\mathcal{H}_{\mathrm{total}}$ of dimension $d_A d_B$ with $d_A, d_B > 1$, there is no unique way to write $\mathcal{H}_{\mathrm{total}} = \mathcal{H}_A \otimes \mathcal{H}_B$ without further physical input. Given any pure state $|\Psi_U\rangle$, choose a unitary U with $U(|0\rangle_A \otimes |0\rangle_B) = |\Psi_U\rangle$ and pull the reference factors through U; in the resulting factorization $|\Psi_U\rangle$ is a product. More generally, observables can be chosen so that a given pure state has any prescribed Schmidt decomposition [8].

What survives is typicality for a family of factorizations fixed in advance.

Theorem A.1 (Entanglement typicality for a specified family)**.** *Let $\mathcal{H}_{\mathrm{total}}$ be finite dimensional and let $\mathfrak{F} = \{\mathbf{f}_1, \ldots, \mathbf{f}_m\}$ be a finite family of nontrivial tensor factorizations selected by a physical rule that does not depend on the sampled state. Write $\mathcal{H}_{\mathrm{total}} = \mathcal{H}_{A_j} \otimes \mathcal{H}_{B_j}$ with $d_{A_j} \le d_{B_j}$. For $\delta > 0$ let*

$$p_j(\delta) := \Pr_{\Psi \sim \mathrm{Haar}} \left[S(\rho_{A_j}) < \log d_{A_j} - \delta \right].$$

Then

$$\Pr\left[S(\rho_{A_j}) \ge \log d_{A_j} - \delta \text{ for every } j \right] \ge 1 - \sum_{j=1}^{m} p_j(\delta),$$

and standard concentration estimates make each $p_j(\delta)$ small at large dimension under the usual conditions on the dimension ratio.

Proof. For each fixed $\mathbf{f}_j$, Page's formula gives [10]

$$\mathbb{E}[S(\rho_{A_j})] = \sum_{k=d_{B_j}+1}^{d_{A_j}d_{B_j}} \frac{1}{k} - \frac{d_{A_j}-1}{2d_{B_j}}, \tag{11}$$

and concentration of measure bounds the lower tail [11]; this tail is $p_j(\delta)$. The union bound over the finite, preselected family proves the inequality. □

The size of the statement matters. For a balanced split $d_A = d_B$, Eq. (11) gives a mean entropy of $\log d_A - \frac{1}{2} + o(1)$, so typical states are close to maximally entangled but keep a gap of about half a nat. Drawing the factorization at random with the state fixed is equivalent to drawing the state at random with the factorization fixed. Hence for a fixed state, product factorizations form a set of measure zero, while factorizations with small but nonzero entanglement have small positive measure.

Theorem A.1 says nothing about all factorizations and does not pick out any member of $\mathfrak{F}$. Which family is admissible, and why, is a question of quantum mereology [12, 13]. Earlier work showed that subsystem structure is induced by the accessible observables rather than given by the Hilbert space [5, 6]. The channel-level results in the main text retain the accessible-observable view while distinguishing its algebra from its resolution.

B Other connectivity constructions

When a metric on a parameter space of coarse-grained states is wanted, the finite-dimensional construction is the quantum Fisher metric. For a smooth faithful family $\lambda \mapsto \rho_{\mathbf{c}}(\lambda)$, define the symmetric logarithmic derivatives by $\partial_a \rho_{\mathbf{c}} = \frac{1}{2}(L_a \rho_{\mathbf{c}} + \rho_{\mathbf{c}} L_a)$ and

$$g^{\mathrm{QF}}_{ab}(\lambda) := \frac{1}{2}\,\mathrm{Tr}[\rho_{\mathbf{c}}(\lambda)\{L_a, L_b\}]\,. \tag{12}$$

This is a positive-semidefinite Riemannian metric on identifiable parameter directions [21]. It is not a Lorentzian spacetime metric, and the pairwise numbers $\mu_{\mathbf{c}}(A, B)$ do not determine it without an additional map from graph or state parameters to geometry.

B.1 Spatial and eigenmode descriptions of a free-fermion chain

Example B.1 (A dimerized free-fermion chain)**.** *Take a periodic chain of $2L$ sites with alternating hopping amplitudes $t_1 = 1$ and $t_2 = 1/2$,*

$$H = -\sum_{c=0}^{L-1} \big(t_1\, c^\dagger_{2c} c_{2c+1} + t_2\, c^\dagger_{2c+1} c_{2c+2} + \mathrm{h.c.}\big),$$

at half filling. The single-particle spectrum has two bands separated by the gap $2|t_1 - t_2| = 1$, and the ground state fills the lower band, $|\Psi_0\rangle = \prod_k a^\dagger_{k,-}|0\rangle$, where $a_{k,-}$ are the lower-band Bloch modes. In the description by occupation numbers of the band-resolved Bloch modes the ground state is a product of filled and empty modes: every set of modes has a pure reduced state, disjoint sets have zero mutual information, and the connectivity graph is empty. In the spatial description blocks of sites are correlated. For $L = 60$ and blocks of four sites, the block entropy saturates at $.3554$ as the block grows, as the area law for gapped one-dimensional systems requires [24, 23]. The mutual information between two blocks falls from $.333$ for adjacent blocks to 9.0×10^{-5} at a separation of ten sites and 5.2×10^{-8} at twenty, close to $e^{-r/1.32}$. The connectivity graph is dominated by edges between neighbouring blocks.

The second description has to be chosen with care. Plain Fourier modes are not enough when the Hamiltonian is not translation invariant: for two disconnected dimers on four sites, $H = -(c^\dagger_0 c_1 + c^\dagger_2 c_3 + \mathrm{h.c.})$ at half filling, the gap at fixed particle number is 2, yet the Fourier modes $k = \pi/2$ and $3\pi/2$ share one particle in superposition and have mutual information $2\ln 2$. A translation-invariant single-band chain with particle conservation is gapless unless its band is full or empty, so a gapped example whose ground state is a product of mode occupations needs band-resolved modes, as above, or in general the single-particle eigenmodes of the Hamiltonian.

The example shows that connectivity depends on the factorization: the spatial description has short-range edges and the eigenmode description has none. It does not exhibit two reconstructed geometries, since its zero-connectivity graph supplies no finite paths under the chosen rule; Proposition 4.3 is the example in which both descriptions produce geometries.

C The nine-probe constant

On the unit sphere, $\sum_k q(r_k)^2$ and Q are quadratic forms in the nine parameters (A_0, b, d), and both are positive definite; the first because the nine probe values determine the parameters, as in the proof of Corollary 3.6. Their generalized eigenvalues do not depend on the basis chosen for the parameters, and a direct computation gives the characteristic polynomial, up to a constant factor,

$$(\lambda - 6)\,(4\lambda^2 - 75\lambda + 90)\,(8\lambda^3 - 225\lambda^2 + 1620\lambda - 2700)^2.$$

Its roots are 1.2886, 2.3962, 6, 7.9001, 17.4614 and 17.8287, the three roots of the cubic each counted twice. The smallest is the smaller root of the quadratic factor, κ_9, and it is attained.

References

[1] W. H. Zurek, *Decoherence, einselection, and the quantum origins of the classical*, Rev. Mod. Phys. **75**, 715 (2003).

[2] J. M. Maldacena, *The Large N limit of superconformal field theories and supergravity*, Int. J. Theor. Phys. **38**, 1113 (1999).

[3] M. Van Raamsdonk, *Building up spacetime with quantum entanglement*, Gen. Relativ. Gravit. **42**, 2323 (2010).

[4] S. Ryu and T. Takayanagi, *Holographic Derivation of Entanglement Entropy from AdS/CFT*, Phys. Rev. Lett. **96**, 181602 (2006).

[5] P. Zanardi, *Virtual Quantum Subsystems*, Phys. Rev. Lett. **87**, 077901 (2001).

[6] P. Zanardi, D. A. Lidar, and S. Lloyd, *Quantum Tensor Product Structures are Observable Induced*, Phys. Rev. Lett. **92**, 060402 (2004).

[7] H. Barnum, E. Knill, G. Ortiz, R. Somma, and L. Viola, *A Subsystem-Independent Generalization of Entanglement*, Phys. Rev. Lett. **92**, 107902 (2004).

[8] N. L. Harshman and K. S. Ranade, *Observables can be tailored to change the entanglement of any pure state*, Phys. Rev. A **84**, 012303 (2011), arXiv:1102.0955.

[9] R. Haag, *Local Quantum Physics: Fields, Particles, Algebras*, Springer-Verlag (1996).

[10] D. N. Page, *Average entropy of a subsystem*, Phys. Rev. Lett. **71**, 1291 (1993).

[11] P. Hayden, D. W. Leung, and A. Winter, *Aspects of generic entanglement*, Commun. Math. Phys. **265**, 95–117 (2006).

[12] J. Cotler, G. Penington, and D. H. Ranard, *Locality from the spectrum*, Commun. Math. Phys. **368**, 1267–1296 (2019).

[13] S. M. Carroll and A. Singh, *Quantum mereology: Factorizing Hilbert space into subsystems with quasiclassical dynamics*, Phys. Rev. A **103**, 022213 (2021).

[14] A. Jenčová, *Comparison of quantum channels and statistical experiments*, arXiv:1512.07016 (2015).

[15] M. B. Ruskai, S. Szarek, and E. Werner, *An analysis of completely-positive trace-preserving maps on $\mathcal{M}_2$*, Linear Algebra Appl. **347**, 159–187 (2002), arXiv:quant-ph/0101003.

[16] A. Fujiwara and P. Algoet, *One-to-one parametrization of quantum channels*, Phys. Rev. A **59**, 3290 (1999).

[17] J. Wallman, C. Granade, R. Harper, and S. T. Flammia, *Estimating the coherence of noise*, New J. Phys. **17**, 113020 (2015), arXiv:1503.07865.

[18] H. Buhrman, R. Cleve, J. Watrous, and R. de Wolf, *Quantum fingerprinting*, Phys. Rev. Lett. **87**, 167902 (2001), arXiv:quant-ph/0102001.

[19] C. W. Helstrom, *Quantum Detection and Estimation Theory*, Academic Press (1976).

[20] W. Hoeffding, *Probability inequalities for sums of bounded random variables*, J. Amer. Statist. Assoc. **58**, 13–30 (1963).

[21] D. Petz, *Monotone metrics on matrix spaces*, Linear Algebra Appl. **244**, 81 (1996).

[22] B. Swingle, *Entanglement Renormalization and Holography*, Phys. Rev. D **86**, 065007 (2012).

[23] J. Eisert, M. Cramer, and M. B. Plenio, *Colloquium: Area laws for the entanglement entropy*, Rev. Mod. Phys. **82**, 277 (2010).

[24] M. B. Hastings, *An area law for one-dimensional quantum systems*, J. Stat. Mech. **2007**, P08024 (2007), arXiv:0705.2024.

[25] S. Liu, *Coupling, Records, and Stable Relations: Formation, Protected Content and the Lifetime of a Geometry*, Zenodo (2026), DOI: 10.5281/zenodo.18367060.

[26] S. Liu, *Measurement as Accessibility: Measurement Permissions, Stable Records and Public Readout*, Zenodo (2026), DOI: 10.5281/zenodo.18400065.

[27] S. Liu, *Renormalization and Imprint Filtering: Source Identification, Memory, and Calibration*, Zenodo (2026), DOI: 10.5281/zenodo.20151451.